

Lightweight Medical Image Segmentation Via Cross-Image Global Structural Filtering Distillation

1st Linxi Tang

*School of Computer Science
Southwest University of Science and Technology
Mianyang, China
1803554228@qq.com*

2nd Wanli Dong

*School of Computer Science
Southwest University of Science and Technology
Mianyang, China
wldong81@foxmail.com*

Abstract—Advances in deep learning have led to highly accurate medical image segmentation models; however, their substantial parameter size poses significant challenges for deployment on edge devices. Knowledge distillation has garnered significant interest as a prominent model compression technique. Nevertheless, existing distillation methods exhibit critical limitations in medical imaging: anatomical structures demonstrate substantial intra-class shape variation and low inter-class contrast, while prevailing techniques overlook global semantic relationships and fail to mitigate feature redundancy—issues that collectively limit distillation performance. To address these gaps, we propose a novel cross-image relational framework, termed Cross-Image Global Structural Filtering Distillation (CIGFD). The framework employs a feature filtering mechanism to eliminate redundancies while capturing essential relational cues at both pixel and region levels, thereby facilitating the efficient transfer of refined global structures across images. Extensive experiments on three public datasets (BTCV, Synapse, and FLAER) demonstrate the consistent and superior performance of CIGFD in boosting the accuracy of compact student models. This work offers a viable pathway for deploying accurate and efficient medical image segmentation models on resource-constrained edge devices.

Index Terms—Cross-Image Structural Relationships, Medical Image Segmentation, Knowledge Distillation, Lightweight Model, Feature Filtering

I. INTRODUCTION

Medical image segmentation is a foundational yet challenging task [1] in medical image analysis, aimed at extracting meaningful regions to facilitate accurate clinical diagnosis and pathological analysis. Deep learning has substantially propelled the advancement of this field. The widespread adoption of U-Net [2] has spurred numerous architectural enhancements to boost segmentation performance. For instance, RA-U-Net [3] incorporates an attention module to accentuate target region features. ResNet34-U-Net [4] replaces U-Net’s original encoder with ResNet34 [5], a design that balances deep feature extraction with the preservation of segmentation details. However, despite these advancements, most state-of-the-art architectures demand substantial computational resources, which hinders their deployment in clinical scenarios. To address these challenges, researchers have devised numerous lightweight models. For example, ENet [6] and ERFNet [7] employ early downsampling and decomposed convolutions to minimize parameters and inference time. ESPNet [8], designed for real-time segmentation, decomposes standard convolutions using pointwise convolutions and a dilated spatial pyramid. A common drawback of such lightweight models, however, is their reliance on operations like pooling and strided convolutions, which reduce feature map resolution. This loss of fine-grained detail often compromises their segmentation accuracy in complex tasks.

Knowledge Distillation (KD) [9] has emerged as a promising technique to enhance the performance of lightweight models. It operates by transferring knowledge from a high-capacity teacher model to a compact student model, enabling the latter to achieve

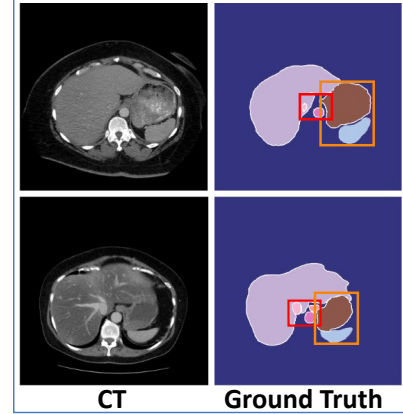


Fig. 1: Visualization of Shape and Texture Heterogeneity. Two representative images from the BTCV dataset are shown, where regions of identical colors correspond to the same organ. Regions within identically-colored bounding boxes exhibit significant morphological and textural heterogeneity, despite belonging to the same organ category.

competitive accuracy with significantly reduced computational costs. Consequently, KD has become a crucial strategy for compressing segmentation models without severely compromising performance [10].

Although existing distillation methods have made progress in segmentation tasks, they generally exhibit several limitations. For instance, the SKD method [11] represents knowledge by constructing structural relationships, IFVD [12] extracts knowledge through semantic masks, and CWD [13] enhances the alignment of salient regions across channels. However, these methods primarily focus on the internal structural information of individual images, often overlooking the importance of global contextual semantics. While EMKD [14] incorporates contrastive learning through regional affinity distillation, its comparative scope remains limited to intra-class comparisons within a single image. When a model relies solely on the internal structural information of individual images, it may result in semantic misalignment between different regions of the same organ, as well as difficulties in distinguishing between organs with similar shapes or locations. This misalignment can hinder the model’s ability to consistently segment various parts of the organ when processing images from different slices or perspectives. Specifically, the lack of global contextual information may prevent the model from maintaining a coherent understanding of the organ’s structure across different slices or viewpoints, leading to inconsistent segmentation results, reduced accuracy, and challenges in distinguishing organs.

Therefore, in medical image segmentation characterized by high variability and significant noise(as shown in Figure 1), there is an urgent need to develop a distillation method capable of precisely filtering and transmitting cross-image global knowledge.

Accordingly, this paper introduces Cross-Image Global Structural Filtering Distillation (CIGFD), a novel distillation framework that extracts pivotal global knowledge through a filtering mechanism. The proposed CIGFD comprises three core modules: (1) Attentional Feature Compaction Module: This module addresses the feature mismatch between teacher and student networks by consolidating multi-channel feature responses into a compact, single-channel representation. This process ensures feature space compatibility and facilitates the transfer of the teacher’s refined feature maps to the student. (2) Feature Filtering Module: To prevent the negative transfer of noisy information prevalent in medical images, this module guides the student to mimic the teacher’s salient features while suppressing ambiguous semantic information, thereby enhancing learning efficiency. (3) Efficient Global Information Module: Contrastive learning based on single images demonstrates limited efficacy in medical imaging, rendering the construction of global structural relationships particularly critical. Leveraging the filtered key features, we establish effective cross-image contrastive learning through corresponding organ masks to learn efficient global structural relationships. The main contributions of this paper are summarized as follows:

- We propose CIGFD, a novel knowledge distillation framework with a feature filtering mechanism for medical image segmentation. It suppresses noise, refines semantics, and enables efficient cross-image structural learning, improving the segmentation efficacy and accuracy of the student model.
- Extensive experiments on three public medical segmentation benchmarks show that CIGFD outperforms current distillation methods, achieving superior accuracy and computational efficiency with lightweight models.

II. METHODOLOGY

We propose the Cross-Image Global Structural Filtering Distillation (CIGFD) framework, which enhances organ discriminability and segmentation completeness in student networks through a feature filtering mechanism. This mechanism captures the feature distributions of organs across different images to facilitate effective knowledge transfer. The CIGFD framework sequentially implements three distillation strategies: (1) acquiring key local details by learning the teacher’s feature compaction maps; (2) transferring critical information via feature filtering from the teacher’s intermediate layers, followed by extracting organ-specific vectors from these filtered features to construct global contextual semantics through cross-image contrastive distillation; and (3) aligning output distributions via logits distillation, allowing the student to learn refined anatomical details from the teacher’s processed outputs.

The logits distillation step [16] follows the foundational work of Hinton et al., where the student mimics the teacher’s output distribution. As this is a well-established technique, we omit a detailed description of its mechanics. The overall architecture of CIGFD is depicted in Figure 2.

A. Attentional feature compaction distillation(AFCD)

To mitigate ineffective learning caused by channel feature mismatch, we introduce an Attentional Feature Compaction (AFC) module based on the channel attention mechanism. This module consolidates the multi-channel features of a layer into an informative feature compaction map. It thereby guides the student to learn

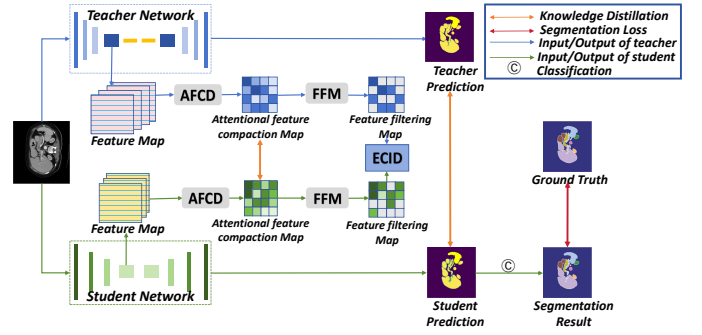


Fig. 2: The overall framework. Our distillation framework integrates four distinct techniques: Attentional feature compaction distillation, Feature filtering module, Effective cross-image contrastive distillation, and Logits Distillation (from left to right).

the teacher’s refined representation of this map, with the specific workflow depicted in Figure 3.

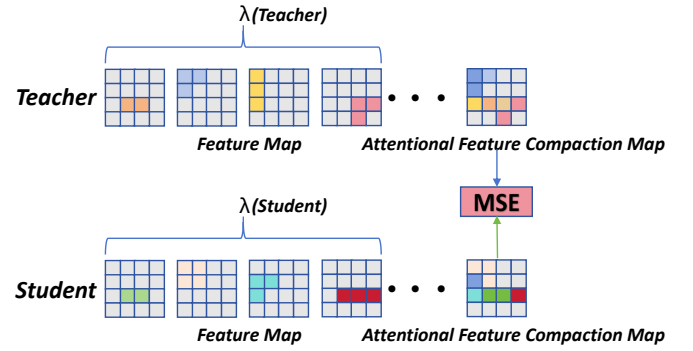


Fig. 3: Attentional Feature Compaction . The intermediate layer features from both teacher and student networks are taken as input. Channel-wise activation weights are then computed and utilized to combine the channel features into a weighted sum. This process guides the student network to learn the teacher’s composite channel characteristics.

First, intermediate features, denoted as $F_t \in \mathbb{R}^{c_t \times w_t \times h_t}$ and $F_s \in \mathbb{R}^{c_s \times w_s \times h_s}$, are extracted from the teacher and student networks, respectively. To align their spatial dimensions, the student feature F_s is scaled to match the spatial size of F_t using a scaling function $f(\cdot)$:

$$F'_s = f(F_s), \quad F'_s \in \mathbb{R}^{c_s \times w_t \times h_t}. \quad (1)$$

Subsequently, channel-wise importance weights are computed. For each channel feature map $e \in \mathbb{R}^{w \times h}$, its global activation response $Z(e)$ is obtained via Global Average Pooling (GAP) applied to its absolute values:

$$Z(e) = \frac{1}{w \times h} \sum_{i=1}^w \sum_{j=1}^h |e(i, j)|, \quad (2)$$

where i and j are spatial indices. The responses for all channels are then normalized into a probability distribution over channels using the softmax function, yielding the channel-wise weights $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_c]$.

Finally, the obtained weights are employed to fuse the multi-channel features into a consolidated feature compaction map. Given an input feature tensor $\varepsilon \in \mathbb{R}^{c \times w \times h}$, where c denotes the total number of channels and $\varepsilon_k \in \mathbb{R}^{w \times h}$ represents the feature matrix of the k -th channel, the feature compaction map is formulated as:

$$\varphi(\varepsilon) = \frac{1}{c} \sum_{k=1}^c \lambda_k \cdot \|\varepsilon_k\|^2 \quad (3)$$

To further transfer the channel feature compaction knowledge, we utilize the Mean Squared Error (MSE) loss to distill the attentional feature compaction map from the teacher to the student. Let φ_t denote the attentional feature compaction map of the teacher network, and φ_s denote that of the student network. The MSE loss is defined as:

$$\mathcal{L}_{\text{AFCD}} = \frac{1}{H \times W} \sum_{h=1}^H \sum_{w=1}^W (\varphi_t(h, w) - \varphi_s(h, w))^2 \quad (4)$$

where H and W are the height and width of the attentional feature compaction map, respectively. Minimizing this loss encourages the student network to generate a compaction map that closely resembles the teacher's, thus effectively distilling the channel-wise feature compaction strategy.

B. Feature filtering module (FFM)

In medical image segmentation, redundant information is far more abundant than effective information. Furthermore, internal organs and soft tissues may exhibit similar features, lacking clear organ discriminability. Thus, constructing an effective global contextual semantic structure is of great significance. However, transferring all the teacher's information to the student would lead to resource waste. To address this, we propose a feature filtering module to establish the foundation for subsequent global structure construction, with detailed architecture illustrated in the FFM component of Figure 2.

First, due to the issue of feature mismatch, we still utilize the previous Attentional Feature Compaction (AFC) module, and this process is defined as:

$$M_t = \varphi(F_t), \quad M_s = \varphi(F_s) \quad (5)$$

Here, M_t and M_s represent the mapped features of the teacher and student networks, respectively, with sizes $w_t \times h_t$. The mapped features combine representations from different channels. In medical image segmentation, it is particularly important for the student network to focus on effective information for segmentation. We specifically screen the foreground activation regions, defined as:

$$p_f(x) = \sigma(x) \quad (6)$$

Here, $p_f(x)$ denotes our foreground activation target, and $\sigma(x)$ represents the sigmoid function, which aims to constrain the score of our foreground target within the range $[0, 1]$. To filter out unimportant features, we introduce a threshold function $f(x)$ to retain important features in the activation map that exceed the threshold, defined as:

$$f(x) = \begin{cases} x, & x \geq \tau \\ 0, & x < \tau \end{cases} \quad (7)$$

Finally, we obtain a filtered feature compaction map, as follows:

$$M_F(e) = f(p_f(M)) \quad (8)$$

C. Effective cross-image contrastive distillation (ECID)

In medical image segmentation, due to differences in the anatomical orientations of images, even the same type of organs exhibit variations in their shapes and textures. Therefore, we argue that the local structures of a single image alone are insufficient to distinguish and completely segment organs. Consequently, we propose cross-image contrastive distillation (ECID) to construct global semantic

structures. However, excessive redundant information may instead lead to negative optimization. Hence, we integrate it with the previously proposed feature filtering module and propose an effective cross-image contrastive distillation method. As shown in Figure 4.

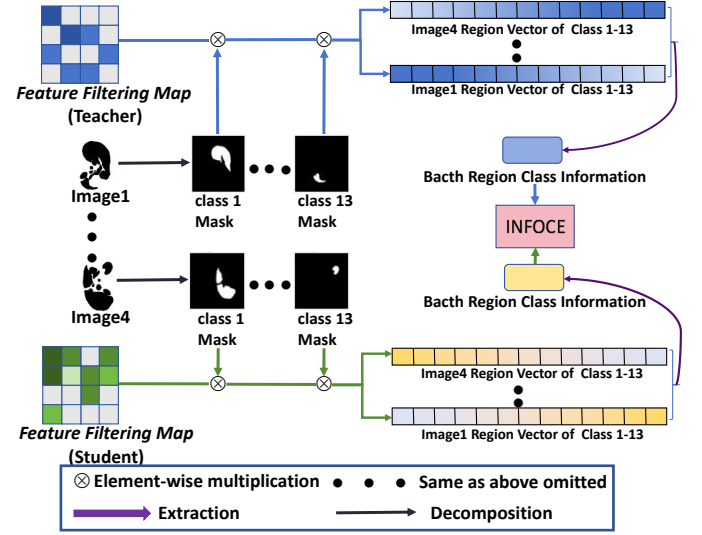


Fig. 4: Effective Cross-image Contrastive Framework. Illustrated with a batch size of 4 on the BTCV dataset, this framework shows how filtered features are combined with organ masks to extract organ-specific vectors. These vectors form a global feature matrix where diagonal positions denote positive sample pairs (same organ) and off-diagonal positions denote negative pairs (different organs). The InfoNCE loss then maximizes similarity along the diagonal while minimizing off-diagonal similarities, thereby enhancing the student network's organ discrimination and segmentation completeness.

We name the filtered teacher and student features M_{Ft} and M_{Fs} respectively. Next, we multiply the filtered feature information by the corresponding extracted mask to extract the organ region mask vector. The formula is as follows:

$$R_c(M_F) = \frac{\sum_{h=1}^H \sum_{w=1}^W (M_F \cdot \hat{m}_c)_{h,w}}{\sum_{h=1}^H \sum_{w=1}^W \hat{m}_c} \quad (9)$$

where $M_F \in \mathbb{R}^{1 \times 1 \times H \times W}$ denotes the filtered feature map, C is the number of class channels, and $\hat{m}_c \in \mathbb{R}^{B \times C \times H \times W}$ represents the class mask for class in the batch. $R_c \in \mathbb{R}^{B \times C \times C_f}$, C_f is the dimension of the feature vector corresponding to the organ. We achieve this by flattening the $B \times C$ dimensions, thereby realizing the function of flattening image features, defined as:

$$V_{\text{vectors}}(M_F) = \text{reshape}(R_c(M_F), [-1, C_f]) \quad (10)$$

Subsequently, the similarity between student and teacher category features is computed as follows:

$$S = V_{\text{vectors}}(M_{Fs}) \cdot V_{\text{vectors}}(M_{Ft})^T \quad (11)$$

where S is the computed similarity matrix, $V_{\text{vectors}}(M_{Fs}) \cdot V_{\text{vectors}}(M_{Ft})^T$ denotes the category similarity using the dot product (the greater the feature similarity, the larger the value), $V_{\text{vectors}}(M_{Ft})^T$ is the transpose of the teacher's feature vector. Finally, the final contrastive loss is computed using the cross-entropy loss. As follows:

$$\mathcal{L}_{\text{ECID}} = \frac{1}{B \times C} \sum_{i=1}^{B \times C} \left(-\log \left(\frac{e^{S_{i,i}/\tau}}{\sum_{j=1}^{B \times C} e^{S_{i,j}/\tau}} \right) \right) \quad (12)$$

TABLE I: Compared to other methods on three datasets.

Methods	Param(M)	FLOPS(G)	FPS(S)	BTCV 512x512 [18]		Synapse 512x512 [19]		Flaer 512x512 [20]	
				Dice(%)	mIoU(%)	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)
T: RA-UNET [2], [3]	22.14	78.22	27.42	86.17	83.11	92.76	90.03	87.00	84.00
S: ENET [6]	0.35	2.01	57.14	83.40	80.03	90.17	87.73	84.75	81.80
CWD [13]	0.35	2.01	57.15	83.67(▲ 0.27)	80.27(▲ 0.24)	88.86(▼ 1.31)	85.10(▼ 2.63)	85.11(▲ 0.36)	81.97(▲ 0.17)
EMKD [14]	0.35	2.01	57.12	84.74(▲ 1.34)	81.61(▲ 1.58)	90.81(▲ 0.64)	87.87(▲ 0.14)	85.88(▲ 1.13)	82.80(▲ 1.00)
CIRKD [15]	0.35	2.01	56.42	84.58(▲ 1.18)	81.35(▲ 1.32)	90.23(▲ 0.06)	87.49(▼ 0.24)	86.27(▲ 1.52)	83.21(▲ 1.41)
BPKD [17]	0.35	2.01	57.31	84.13(▲ 0.73)	80.64(▲ 0.61)	89.01(▼ 1.16)	85.56(▼ 2.17)	86.24(▲ 1.49)	82.96(▲ 1.16)
CIGFD(Ours)	0.35	2.01	61.24	85.46(▲ 2.06)	82.42(▲ 2.39)	92.13(▲ 1.96)	89.17(▲ 1.44)	86.98(▲ 2.23)	83.97(▲ 2.17)
T: RA-UNET [2], [3]	22.14	78.22	27.42	86.17	83.11	92.76	90.03	87.00	84.00
S: RESNET18 [5]	11.18	9.52	47.62	77.92	73.84	89.93	85.56	81.35	76.58
CWD [13]	11.18	9.52	47.91	78.04(▲ 0.12)	73.95(▲ 0.11)	87.55(▼ 2.38)	82.64(▼ 2.92)	81.89(▲ 0.54)	76.71(▲ 0.13)
EMKD [14]	11.18	9.52	47.12	80.60(▲ 2.68)	76.47(▲ 2.63)	89.42(▼ 0.51)	84.98(▼ 0.58)	82.79(▲ 1.44)	77.95(▲ 1.37)
CIRKD [15]	11.18	9.52	46.62	80.35(▲ 2.43)	76.29(▲ 2.45)	90.37(▲ 0.44)	86.13(▲ 0.57)	82.17(▲ 0.82)	77.72(▲ 1.14)
BPKD [17]	11.18	9.52	47.86	78.85(▲ 0.93)	74.60(▲ 0.76)	89.27(▼ 0.66)	84.59(▼ 0.97)	82.33(▲ 0.98)	77.43(▲ 0.85)
CIGFD(Ours)	11.18	9.52	49.15	80.73(▲ 2.81)	76.31(▲ 2.47)	90.38(▲ 0.45)	86.15(▲ 0.59)	83.02(▲ 1.67)	78.31(▲ 1.73)
T: RESNET34-UNET [4]	16.30	94.78	26.34	87.09	84.02	91.62	88.83	85.61	82.71
S: ENET [6]	0.35	2.01	57.14	83.40	80.03	90.17	87.73	84.75	81.80
CWD [13]	0.35	2.01	53.81	81.62(▼ 1.78)	77.96(▼ 2.07)	89.70(▼ 0.47)	86.19(▼ 1.54)	84.91(▲ 0.16)	81.97(▲ 0.17)
EMKD [14]	0.35	2.01	60.12	84.14(▲ 0.74)	80.92(▲ 0.89)	89.63(▼ 0.54)	86.64(▼ 1.09)	85.80(▲ 1.05)	82.37(▲ 0.57)
CIRKD [15]	0.35	2.01	57.46	84.54(▲ 1.14)	81.06(▲ 1.03)	90.92(▲ 0.75)	87.85(▲ 0.12)	85.28(▲ 0.53)	82.15(▲ 0.35)
BPKD [17]	0.35	2.01	56.21	82.96(▼ 0.44)	79.33(▼ 0.70)	89.11(▼ 1.06)	86.76(▼ 0.97)	84.97(▲ 0.22)	81.96(▲ 0.16)
CIGFD(Ours)	0.35	2.01	61.54	85.57(▲ 2.17)	82.23(▲ 2.20)	93.53(▲ 3.36)	90.89(▲ 3.16)	87.00(▲ 2.25)	83.77(▲ 1.97)
T: RESNET34-UNET [4]	16.30	94.78	26.34	87.09	84.02	91.62	88.83	85.61	82.71
S: RESNET18 [5]	11.18	9.52	47.62	77.92	73.84	89.93	85.56	81.35	76.58
CWD [13]	11.18	9.52	47.12	78.85(▲ 0.93)	74.87(▲ 1.03)	86.39(▼ 3.54)	81.49(▼ 4.07)	81.78(▲ 0.43)	76.76(▲ 0.18)
EMKD [14]	11.18	9.52	46.23	80.14(▲ 2.22)	76.02(▲ 2.18)	89.25(▼ 0.68)	84.76(▼ 0.8)	82.68(▲ 1.33)	78.02(▲ 1.44)
CIRKD [15]	11.18	9.52	46.57	78.77(▲ 0.85)	74.75(▲ 0.91)	89.81(▼ 0.12)	85.50(▼ 0.06)	81.80(▲ 0.45)	77.18(▲ 0.60)
BPKD [17]	11.18	9.52	45.17	78.45(▲ 0.53)	74.20(▲ 0.36)	89.77(▼ 0.16)	85.09(▼ 0.47)	82.00(▲ 0.65)	77.28(▲ 0.70)
CIGFD(Ours)	11.18	9.52	48.46	80.01(▲ 2.09)	76.06(▲ 2.22)	90.22(▲ 0.29)	85.90(▲ 0.34)	82.73(▲ 1.38)	78.12(▲ 1.54)

where τ denotes the temperature parameter that regulates the smoothness of the similarity distribution. The diagonal elements $S_{i,i}$ of the matrix S represent the similarity between student and teacher features of the same class within the batch, while the off-diagonal elements $S_{i,j}$ correspond to the similarity between features of different classes. After softmax normalization, the similarity of same-class features is amplified, guiding the student model to align with the teacher model. Conversely, the similarity between different-class features is relatively low and further suppressed by the softmax, enhancing the discriminability between classes. Employing cross-entropy loss effectively aligns same-class features while significantly increasing the distinction among different-class features.

III. EXPERIMENT

A. Experimental setup

To comprehensively evaluate the proposed Cross-Image Global Structural Filtering Distillation (CIGFD) framework, we employ ResNet-34 UNet [4] and RA-UNet as teacher networks, while utilizing ENet [6] and ResNet-18 [5] as lightweight student networks. All networks are optimized using Adam with an initial learning rate of 1×10^{-3} and trained for 50 epochs with a batch size of 4. We adopt a cosine annealing schedule to decay the learning rate to a minimum of 1×10^{-6} . To enhance generalization capability, data augmentation techniques including random scaling and rotation are applied during training. Extensive experiments are conducted on three medical image segmentation benchmarks: BTCV (13 organs) [18], Synapse (8 organs) [19], and Flaer (4 organs) [20]. All experiments are implemented on an NVIDIA RTX 4090 GPU with 24 GB memory.

B. Evaluation metric

In medical image segmentation, we evaluated using Dice and mIoU. The Dice coefficient is defined as:

$$\text{Dice}(P, G) = \frac{2 \sum_{c=1}^C |P_c \cap G_c|}{\sum_{c=1}^C |P_c| + |G_c|} \quad (13)$$

The mIoU metric is defined as:

$$\text{mIoU}(P, G) = \frac{1}{C} \sum_{c=1}^C \frac{|P_c \cap G_c|}{|P_c \cup G_c|} \quad (14)$$

where C represents the number of classes, P_c and G_c represent the predicted and ground truth regions of class c , respectively. To ensure fairness, we computed the average of the sum of Dice and mIoU in all classes as the performance metric.

C. Comparative experiments

To validate the superiority of our proposed method, we conducted comprehensive comparisons with various state-of-the-art knowledge distillation approaches. The compared methods include both general-purpose distillation frameworks such as CWD [13], and specialized distillation techniques designed for semantic segmentation, including EMKD [14], CIRKD [15], and BPKD [17]. To thoroughly validate the architectural independence of our method, we selected two distinct teacher networks [3] [4] and two different student networks [5] [6], constructing multiple teacher-student architecture combinations for evaluation.

Table 1 presents comparative experimental results across three distinct datasets, convincingly demonstrating the significant advantages of our proposed method. Across all evaluated models, CIGFD consistently achieves more substantial performance improvements compared to previous approaches. Our method improved the student

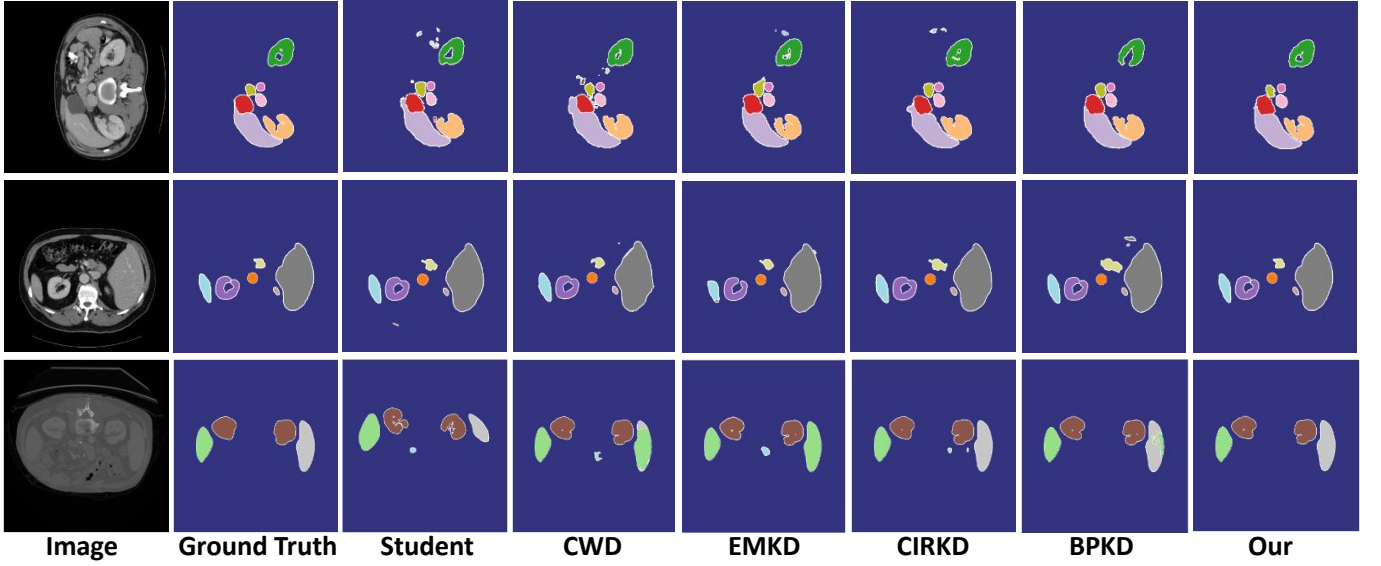


Fig. 5: Results are organized by dataset (Teacher: RA-UNet, Student: ENet): Row 1 shows BTCV results, Row 2 displays Synapse results, and Row 3 presents FLAER results.

TABLE II: Ablation experiments on three datasets.

Method	BTCV [18]		Synapse [19]		Flaer [20]	
	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)	Dice(%)	mIoU(%)
Teacher: RA-UNET [2], [3]	86.17	83.11	92.76	90.03	87.00	84.00
Student: ENET [6]	83.40	80.03	90.17	87.73	84.75	81.80
+ LD	84.05(▲ 0.65)	80.45(▲ 0.42)	90.31(▲ 0.14)	87.81(▲ 0.08)	84.87(▲ 0.12)	81.94(▲ 0.14)
+ AFCD	84.47(▲ 1.07)	81.27(▲ 1.24)	90.83(▲ 0.66)	88.01(▲ 0.28)	85.18(▲ 0.43)	82.14(▲ 0.34)
+ ECID	84.71(▲ 1.31)	81.75(▲ 1.72)	90.95(▲ 0.78)	88.35(▲ 0.62)	85.46(▲ 0.71)	82.32(▲ 0.52)
+ AFCD + LD	84.87(▲ 1.47)	81.91(▲ 1.88)	91.19(▲ 1.02)	88.98(▲ 1.25)	85.69(▲ 0.94)	82.72(▲ 0.92)
+ ECID + LD	84.99(▲ 1.59)	81.71(▲ 1.68)	90.79(▲ 0.62)	88.47(▲ 0.74)	85.73(▲ 0.98)	82.43(▲ 0.63)
+ AFCD + ECID + LD	85.46(▲ 2.06)	82.42(▲ 2.39)	92.13(▲ 1.96)	89.17(▲ 1.44)	86.98(▲ 2.23)	83.97(▲ 2.17)

model's Dice and mIoU by at least 0.29%, with the maximum improvement reaching 3.36% and 3.16%, respectively. Specifically, on the BTCV [18] dataset, when employing RESNET34-UNET [4] as the teacher model and ENET [6] as the student model, both CWD [13] and BPKD [17] methods exhibit noticeable performance degradation. Similarly, on the Synapse [19] dataset under configurations where RA-UNET [3] and RESNET34-UNET [4] serve as teacher networks while ENET [6] and RESNET18 [5] act as student networks, all existing comparative methods show varying degrees of performance decline. These results strongly indicate that our method maintains excellent generalization capability and architectural independence across different model architectures and datasets.

It is particularly noteworthy that the incorporation of our feature filtering mechanism brings remarkable improvement in training efficiency. As evidenced by the bolded FPS values in Table 1, our approach demonstrates substantially higher processing speed compared to existing techniques, further highlighting its practical deployment value. Figure 5 presents segmentation results by our method on the BTCV [18], Synapse [19], and Flaer [20] datasets. Evidently, our segmentation results exhibit greater completeness with finer details such as sharper boundaries, closely approximating the ground truth. This demonstrates that our distillation approach enables the student network to effectively acquire both structural and detailed knowledge, thereby achieving superior segmentation performance.

D. Ablation study

In this section, we conducted a systematic ablation study to comprehensively evaluate individual components of our method. All experiments were performed on three datasets using a consistent configuration with T: RA-UNET and S: ENET architectures. As shown in Table 2, we sequentially assessed the contributions of Logits Distillation (LD) [16], Attentional feature compaction distillation (AFCD), and Effective cross-image contrastive distillation (ECID).

We use the BTCV dataset as a representative case for detailed analysis, as similar trends are observed across all datasets. When applying LD alone, although BTCV dataset show improvement, the performance gain remains limited (maximum of only 0.65%), indicating that relying solely on local detail information is insufficient for significant performance breakthroughs. In contrast, introducing AFCD yields substantial performance improvement (up to 1.07%), demonstrating that the feature compaction mechanism effectively enhances the transmission efficiency of critical information. Similarly, ECID also exhibits remarkable effectiveness (maximum gain of 1.31%), validating the important role of cross-image contrastive learning in enhancing feature discriminability.

Notably, when integrating multiple methods, we observe clear synergistic effects: the combination of AFCD and LD achieves a 1.47% improvement, ECID with LD reaches 1.59% gain, and the integration

of all three components obtains the maximum improvement of 2.06%, fully demonstrating the complementary nature of these components. This progressive performance enhancement verifies the rationality of our design philosophy: local detail preservation, feature compaction enhancement, and cross-image contrastive learning collectively form a complete and effective distillation framework.

E. Overall loss

We combine the modules mentioned above to train the student network using the formula shown in the end. The total loss function is as follows:

$$L_{\text{total}} = L_{\text{SEG}} + \alpha_1 L_{\text{AFCD}} + \alpha_2 L_{\text{ECID}} + \beta L_{\text{LD}} \quad (15)$$

where L_{seg} is the general cross-entropy loss function. The hyperparameters α_1 , α_2 , and β are set to 0.5, 0.1, and 0.2, respectively. The threshold τ for feature filtering is set to 0.5 in our design

F. Hyperparameter Sensitivity Analysis

In our experiments, we independently tuned all hyperparameters and found that α_1 , α_2 , and β had minimal impact on model performance, indicating the robustness of our method to these parameters. In contrast, the feature filtering threshold τ had a significant impact on model behavior.

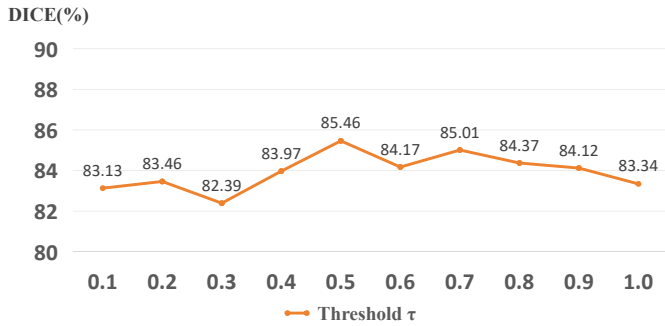


Fig. 6: Threshold τ Sensitivity for Feature Filtering.

When τ is set to a high value, only highly salient feature responses are retained. While this effectively filters out noise, it may also lead to the loss of critical semantic information, thereby impairing the model's ability to delineate fine-grained anatomical structures. Conversely, a lower τ preserves a large number of non-salient features, which helps retain contextual information but introduces redundant noise, reducing the model's attention to salient features.

Based on our experiments, we recommend setting τ within the range of [0.4, 0.6] to balance detail preservation and noise suppression (see Figure 6).

IV. CONCLUSION

This paper addresses the challenges of class distinction difficulties and segmentation incompleteness in medical image segmentation caused by organ shape diversity, proposing two novel distillation methods: Attentional feature compaction distillation (AFCD) and Effective cross-image contrastive distillation (ECID). AFCD enhances the model's ability to capture crucial semantic features by extracting and compressing multi-scale feature maps from intermediate layers, while ECID promotes the learning of more discriminative feature representations through establishing effective cross-image contrastive learning. Extensive experiments on three public medical imaging datasets demonstrate that our proposed CIGFD framework not only

significantly improves segmentation accuracy but also offers advantages of training efficiency and fast inference speed, exhibiting excellent potential for practical deployment.

REFERENCES

- [1] Wang R, Lei T, Cui R, et al. Medical image segmentation using deep learning: A survey[J]. IET image processing, 2022, 16(5): 1243-1267.
- [2] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in Int. Conf. Med. Image Comput. Comput.-Assist. Intervent., 2015, pp. 234-241.
- [3] Z. Li, H. Zhang, Z. Li, and Z. Ren, "Residual-attention UNet++: A nested residual-attention U-net for medical image segmentation," Appl. Sci., vol. 12, no. 14, p. 7149, 2022.
- [4] M. Z. Alom, C. Yakopcic, M. Hasan, T. M. Taha, and V. K. Asari, "Recurrent residual U-Net for medical image segmentation," J. Med. Imag., vol. 6, no. 1, p. 014006, 2019.
- [5] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 770-778.
- [6] A. Paszke, A. Chaurasia, S. Kim, and E. Culurciello, "ENet: A deep neural network architecture for real-time semantic segmentation," arXiv preprint arXiv:1606.02147, 2016.
- [7] Romera E, Alvarez J M, Bergasa L M, et al. Erfnet: Efficient residual factorized convnet for real-time semantic segmentation[J]. IEEE Transactions on Intelligent Transportation Systems, 2017, 19(1): 263-272.
- [8] Watanabe S, Hori T, Karita S, et al. ESPnet: End-to-end speech processing toolkit[J]. arXiv preprint arXiv:1804.00015, 2018.
- [9] Gou J, Yu B, Maybank S J, et al. Knowledge distillation: A survey[J]. International journal of computer vision, 2021, 129(6): 1789-1819.
- [10] Zhang Y, Shen Z, Jiao R. Segment anything model for medical image segmentation: Current applications and future directions[J]. Computers in Biology and Medicine, 2024, 171: 108238.
- [11] Liu Y, Chen K, Liu C, et al. Structured knowledge distillation for semantic segmentation[C]//Proceedings of the IEEE/CVF conference on computer vision and pattern recognition. 2019: 2604-2613.
- [12] Y. Wang, W. Zhou, T. Jiang, X. Bai, and Y. Xu, "Intra-class feature variation distillation for semantic segmentation," in Eur. Conf. Comput. Vis., 2020, pp. 346-362.
- [13] C. Shu, Y. Liu, J. Gao, Z. Yan, and C. Shen, "Channel-wise knowledge distillation for dense prediction," in Proc. IEEE/CVF Int. Conf. Comput. Vis., 2021, pp. 5311-5320.
- [14] D. Qin, J.-J. Bu, Z. Liu, X. Shen, S. Zhou, J.-J. Gu, Z.-H. Wang, L. Wu, and H.-F. Dai, "Efficient medical image segmentation based on knowledge distillation," IEEE Trans. Med. Imag., vol. 40, no. 12, pp. 3820-3831, 2021.
- [15] C. Yang, H. Zhou, Z. An, X. Jiang, Y. Xu, and Q. Zhang, "Cross-image relational knowledge distillation for semantic segmentation," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 12319-12328.
- [16] Hinton G, Vinyals O, Dean J. Distilling the knowledge in a neural network[J]. arXiv preprint arXiv:1503.02531, 2015.
- [17] L. Liu, Z. Wang, M. H. Phan, B. Zhang, J. Ge, and Y. Liu, "BPKD: Boundary privileged knowledge distillation for semantic segmentation," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis., 2024, pp. 1062-1072.
- [18] B. Landman, Z. Xu, J. Igelsias, M. Styner, T. Langerak, and A. Klein, "MICCAI multi-atlas labeling beyond the cranial vault—workshop and challenge," in Proc. MICCAI Multi-Atlas Labeling Beyond Cranial Vault Workshop Challenge, vol. 5, 2015, p. 12.
- [19] S. K. Zhou, R. Green, et al., "The Synapse Challenge: A Large-Scale Benchmark for 3D Medical Image Segmentation," in MICCAI, 2015. [Online]. Available: <https://www.synapse.org/>
- [20] J. Ma, Y. Zhang, S. Gu, X. An, Z. Wang, C. Ge, C. Wang, F. Zhang, Y. Wang, Y. Xu, S. Gou, F. Thaler, C. Payer, D. Štern, E. G. A. Henderson, D. M. McSweeney, A. Green, P. Jackson, L. McIntosh, Q.-C. Nguyen, A. Qayyum, P.-H. Conze, Z. Huang, Z. Zhou, D.-P. Fan, H. Xiong, G. Dong, Q. Zhu, J. He, and X. Yang, "Fast and Low-GPU-memory abdomen CT organ segmentation: The FLARE challenge," Med. Image Anal., vol. 82, p. 102616, 2022.